

Education

Monash University

*Department of Econometrics and
Business Statistics*

Doctor of Philosophy

Jan 2023 - present

- Thesis title: "Investigations of Flexible Jump Modelling and Systematic Risk Implied by Option Information"

The University of New South Wales

Department of Finance

Visiting PhD

Oct, 2024 - Dec, 2024

- Hosted by Andrew Patton

University of Melbourne

Melbourne Business School

Doctor of Philosophy

July 2022 - Jan 2023

- I started my PhD at Melbourne Business School and then transferred to Monash University to continue my PhD degree.

Monash University

Monash Business School

Bachelor of Commerce (1st Class Honors)

Feb 2017 - Dec 2020

- Major in Finance and Econometrics
- Graduated with High Distinction
- GPA: 3.9/4.0

Research Projects

Downside and Upside Systematic Risk Measure Implied by Option Information

Monash University

Nov 2024 - Present

- **Abstract:** We propose an innovative method of constructing the option implied downside and upside systematic risk measure through option implied truncated moments. A by-product of the proposed method is the option implied idiosyncratic risk of equity. We investigate the empirical application of these proposed risk measures in interpreting and predicting the return behaviours under physical measure.

Learning Financial Jumps with Infinite Gaussian Mixture

Monash University & University of Melbourne

June 2022 - Present

- **Abstract:** We propose a Bayesian nonparametric method using Dirichlet process mixture to model jumps. This new model encompasses existing parametric jump specification, but allows for extreme jumps to occur in either tail of the return distribution via infinite mixture model. The flexibility in our model specification allows for multi-modal jump size distribution. We investigate the robustness of our proposed model under different jump intensity and size specifications under simulations and provide comparisons about the model-driven option implied volatility smiles. We verify the practical use of our model with empirical data in jump detection, data-driven jump size density and implied volatility smiles.

Optimal probabilistic forecasts for risk management

Monash University

Mar 2020 - Mar 2023

- **Abstract:** This paper explores the implications of producing forecast distributions that are optimized according to scoring rules that are relevant to financial risk management. We assess the predictive performance of optimal forecasts from potentially misspecified models for i) value-at-risk and expected shortfall predictions; and ii) prediction of the VIX volatility index for use in hedging strategies involving VIX futures. Our empirical results show that calibrating the predictive distribution using a score that rewards the accurate prediction of extreme returns improves the VaR and ES predictions. Tail-focused predictive distributions are also shown to yield better outcomes in hedging strategies using VIX futures.
- **Link to the paper:** <https://arxiv.org/abs/2303.01651>

Conference Presentations

The European Seminar on Bayesian Econometrics (ESOBE 2025)

Melbourne Business School, Australia

August, 2025

- Poster presentation

The 8th International Conference on Econometrics and Statistics (EcoSta 2025)

Waseda University, Japan

August, 2025

- Invited Talk

World Congress of the Econometric Society

Seoul, South Korea

August, 2025

Annual Conference of the International Association for Applied Econometrics University of Turin, Italy	June, 2025
The 19th International Symposium on Econometric Theory and Applications: SETA 2025 University of Macau, China Submission Accepted	June, 2025
Econometric Society Australian Meetings (ESAM) Monash University, Australia	December, 2024
Time Series and Forecasting Symposium University of Sydney, Australia • Best Poster Runner-up	November, 2024

Work Experience

Monash Business School Teaching Associate • Taught undergraduate and postgraduate courses in econometrics, including linear regression, ARIMA time-series forecasting, and principal component analysis (PCA). • Programming using R to demonstrate forecasting, modelling, data analysis and interpretation techniques. • Guided students in applying quantitative models to real-world financial and economic data, enhancing their analytical and presentation skills.	Melbourne, Australia Feb 2023 - present
Monash Business School Research Assistant • Processed and analysed high-frequency bond data for the Thailand Bond Market Association using R; • Evaluated the price jumps following regulation changes.	Melbourne, Australia Aug 2023 - Oct 2024
Melbourne Business School Research Assistant • Extracted and processed high-frequency equity data from Refinitiv for all S&P 500 constituents. • Automated large-scale data cleaning and transformation in R.	Melbourne, Australia Aug 2022 - Jan 2023
Liaowang Think Tank - Xinhua News Agency Research Analyst • Collaborated with a small team for economics research. Provide assistance to the bigger research team about background analysis. • Drafted analysis reports and policy recommendations from the government’s perspective.	Beijing, China July 2021 - Oct 2021

Research Grants

Prestigious International Conference Award Monash Business School, 2025	2,300 AUD
CFM/SoFiE Scholarship Capital Fund Management, The Society for Financial Econometrics, 2025	Conference Accommodation Support
Departmental Funding for External Research Experience (Visiting Scholar Support) Department of Economics and Business Statistics at Monash University, 2024	6,400 AUD

Honors & Awards

2024	Excellent Teaching Team , Monash Business School
2024	Best Poster Runner Up , Time Series and Forecasting Symposium, University of Sydney
2023	Co-funded Monash Graduate Scholarship , Monash University
2023	Monash Business School Top-Up Scholarship , Monash Business School
2022	Faculty of Business and Economics Graduate Research Scholarship , University of Melbourne
2021	2020 Dean’s Honor (First Class) , Monash Business School
2020	Econometrics Memorial Scholarship , Department of Econometrics and Business Statistics, Monash
2020	Meritorious Winner , 2020 Interdisciplinary contest in Modelling